



BUSINESS INTELLIGENCE AND MANAGEMENT INTEGRATION

A CASE STUDY OF PIMS HOSPITAL



Executive summary

This project evaluates how Business Intelligence (BI) can improve hospital management at PIMS.

Although PIMS has initiated digital systems, only 6 out of 28 modules are functional, creating major operational gaps.

Using Power BI, we analyze hospital operations, inventory risks, and strategic challenges to support data-driven decision making.

ORGANIZATION OVERVIEW



NATIONAL LEVEL HEALTHCARE INSTITUTE

PIMS is a federally administered tertiary care hospital providing specialized and emergency healthcare services across Pakistan.



WIDE SERVICE COVERAGE

High patient inflow due to national importance and affordability places constant pressure on hospital operations.



PUBLIC SECTOR GOVERNANCE STRUCTURE

It operates under the Ministry of National Health Services, ensuring regulatory compliance and public-sector accountability.



OPERATIONAL & DIGITAL COMPLEXITY

The scale of services requires efficient management systems for transparency, resource optimization, and decision-making.



PROBLEM STATEMENT

- ▶ Systems at PIMS are fragmented.
▶ Departments do not share data properly.
- ▶ Management lacks real-time visibility.
▶ Patient safety is at risk.
- ▶ Manual work causes:
 - Delays
 - Errors
 - Medicine shortages

DIGITAL GAP PROBLEM

PIMS planned 28 digital hospital modules, but only 6 are currently operational.

The remaining modules are disconnected due to coordination and governance issues.

This gap results in fragmented data, manual workarounds, and lack of centralized visibility for management.

- ✓ INTEROPERABILITY ISSUES
- ✓ OUTDATED SYSTEMS
- ✓ DATA SILOS



Administrative Structure

- ▶ Structure supports control but slows digital coordination.
- ▶ PIMS follows a formal management hierarchy
- ▶ Main roles:
 - Board of Governors
 - Executive Director
 - Joint Executive Directors
 - Medical, Hospital & Nursing Directors

Managerial Decision Making

- ▶ **Strategic level:** Focuses on long-term planning and overall organizational goals.
- ▶ **Tactical level:** Deals with efficient allocation and management of resources.
- ▶ **Operational level:** Involves day-to-day patient care and routine activities.
- ▶ **Current approach:** Management is mostly reactive in nature.
- ▶ **Decision-making:** Actions are taken after issues or shortages occur.
- ▶ **Role of BI:** Business Intelligence enables a shift toward proactive, data-driven management.

KEY ORGANIZATIONAL PROCESSES

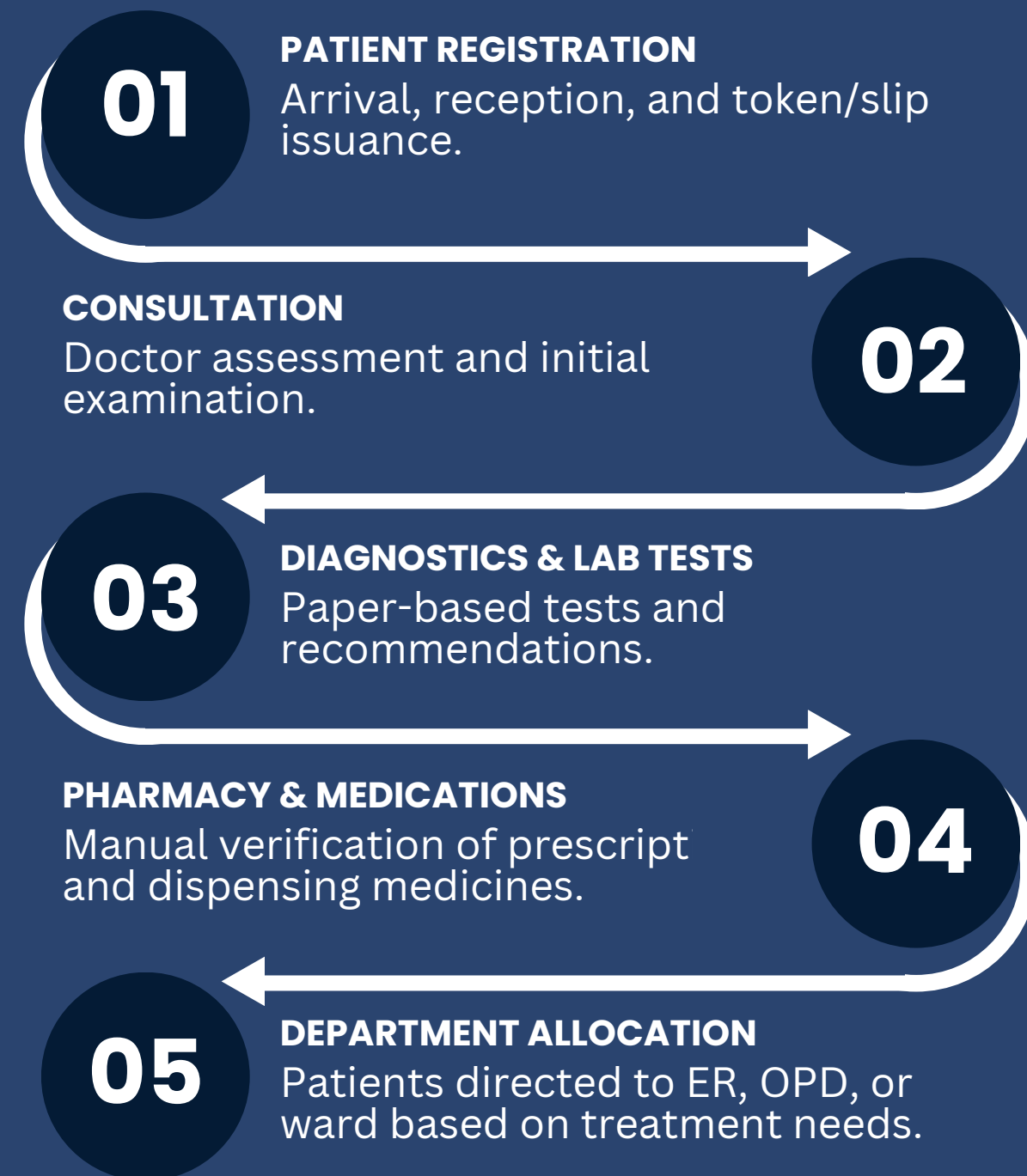
PIMS operations are driven by interconnected clinical and support processes. These processes ensure efficient healthcare delivery and patient safety. Effective coordination among them is essential for hospital performance.



Handles patient admissions, triage, and emergency response to ensure timely care.

Manages procurement, storage, and availability of medicines to avoid shortages.

Supports connected medical devices while protecting patient data and systems.



CURRENT PATIENT FLOW

Patients first report to reception for registration and receive a manual slip. They physically move between wards, doctors, pharmacy, and laboratory using paper vouchers.

Because systems are not integrated, patients themselves carry medical information, causing delays and errors

ROLE OF BUSINESS INTELLIGENCE (BI)

- ▶ Microsoft Power BI is used
- ▶ BI converts data into visual dashboards
- ▶ Helps management:
 - Monitor operations
 - Control inventory
 - Identify risks early
- ▶ Supports data-driven decision making

BI-DASHBOARD

HOSPITAL OPERATIONS

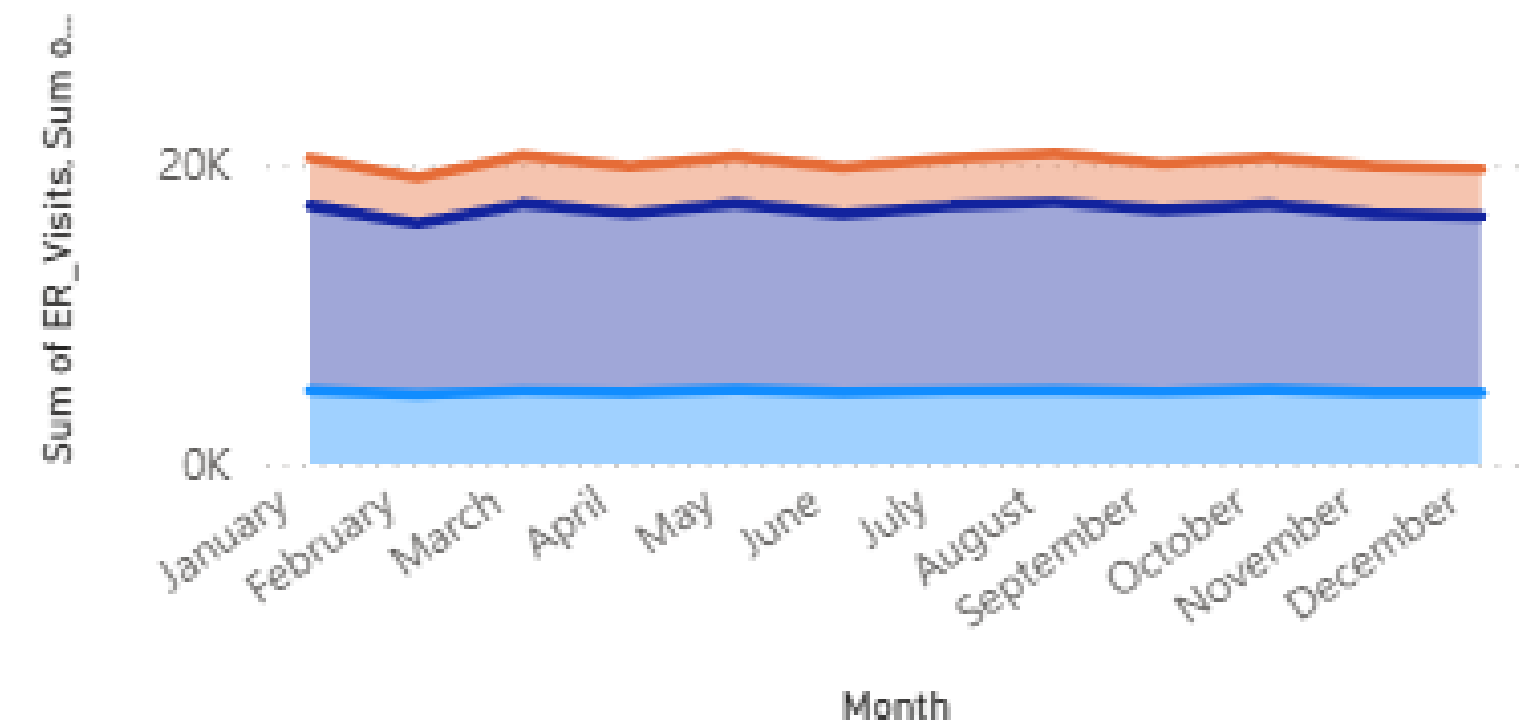
Power BI dashboards track ER and OPD visits, showing an average of ~20,000 patients per month.

The dashboard also highlights staffing and bed occupancy trends.

A key finding was a staffing dip in February, despite steady patient inflow, indicating resource imbalance.

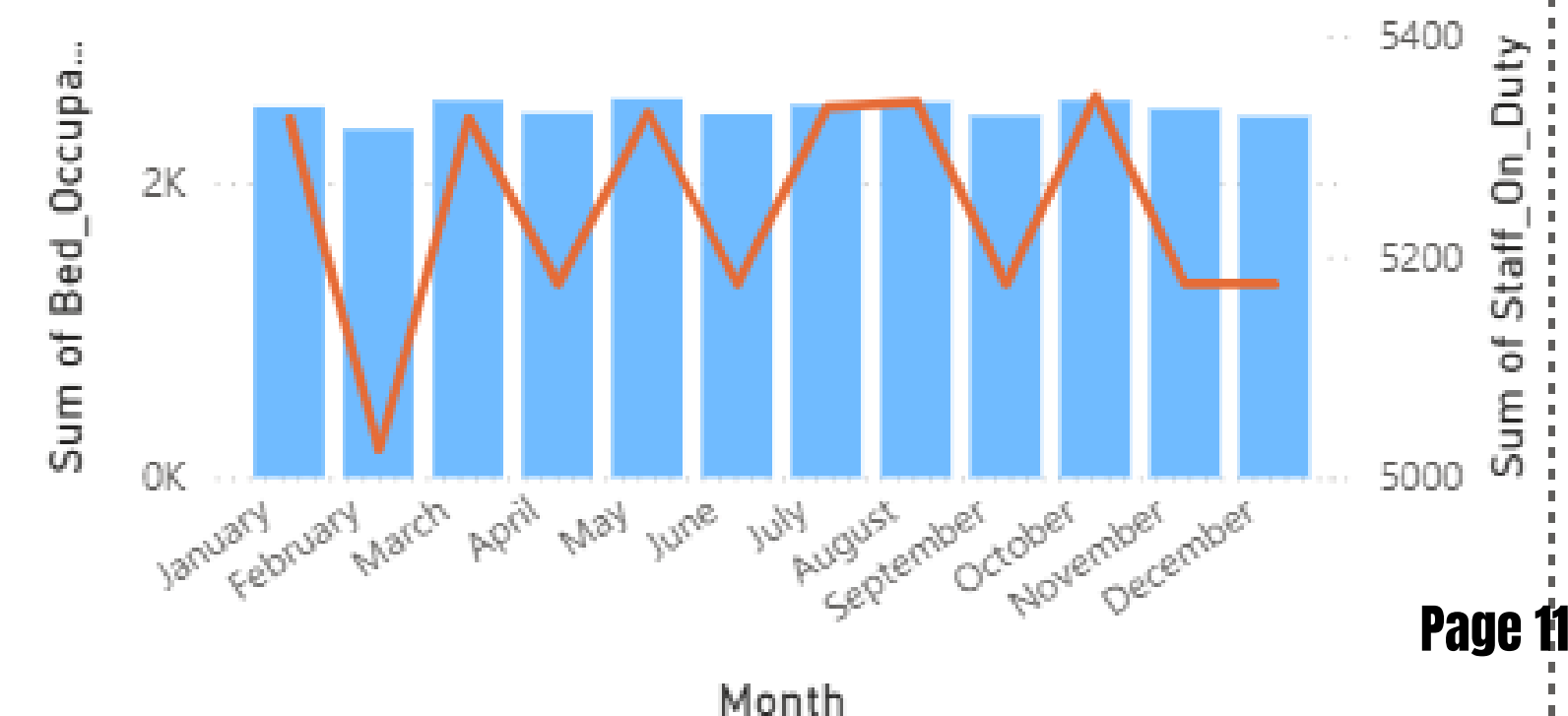
Sum of ER_Visits, Sum of OPD_Visits and Sum of Total_Admissions by Month

● Sum of ER_Visits ● Sum of OPD_Visits ● Sum of Total_Admissions



Resource Balance: Bed Occupancy vs. Staffing Trends

● Sum of Bed_Occupancy_Rate ● Sum of Staff_On_Duty



BI-DASHBOARD

INVENTORY MANAGEMENT DASHBOARD

- ▶ Tracks medicine stock in real time
- ▶ Sends alerts before stock-out
- ▶ Focus on life-saving medicines like insulin
- ▶ Improves patient safety

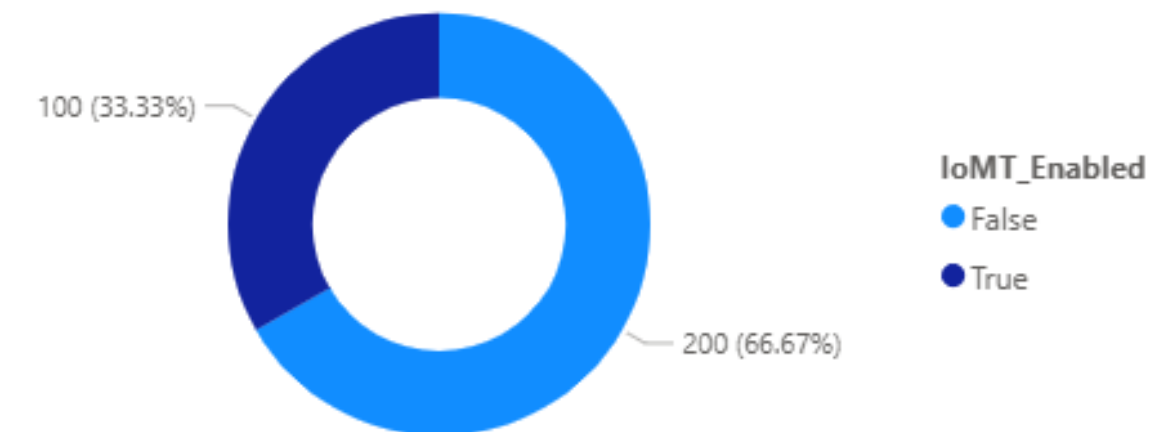
Medicine_Name	Sum of Stock_Quantity	Sum of Reorder_Level
Azithromycin 2200mg	892	78
Azithromycin 2300mg	447	73
Azithromycin 2400mg	1002	68
Azithromycin 2500mg	557	63
Azithromycin 2600mg	112	58
Azithromycin 2700mg	667	53
Azithromycin 2800mg	222	48
Azithromycin 2900mg	777	43
Azithromycin 3000mg	332	38
Azithromycin 3100mg	887	33
Azithromycin 3200mg	447	28
Total	275750	59350

BI-DASHBOARD

IOMT & RISK DASHBOARDS

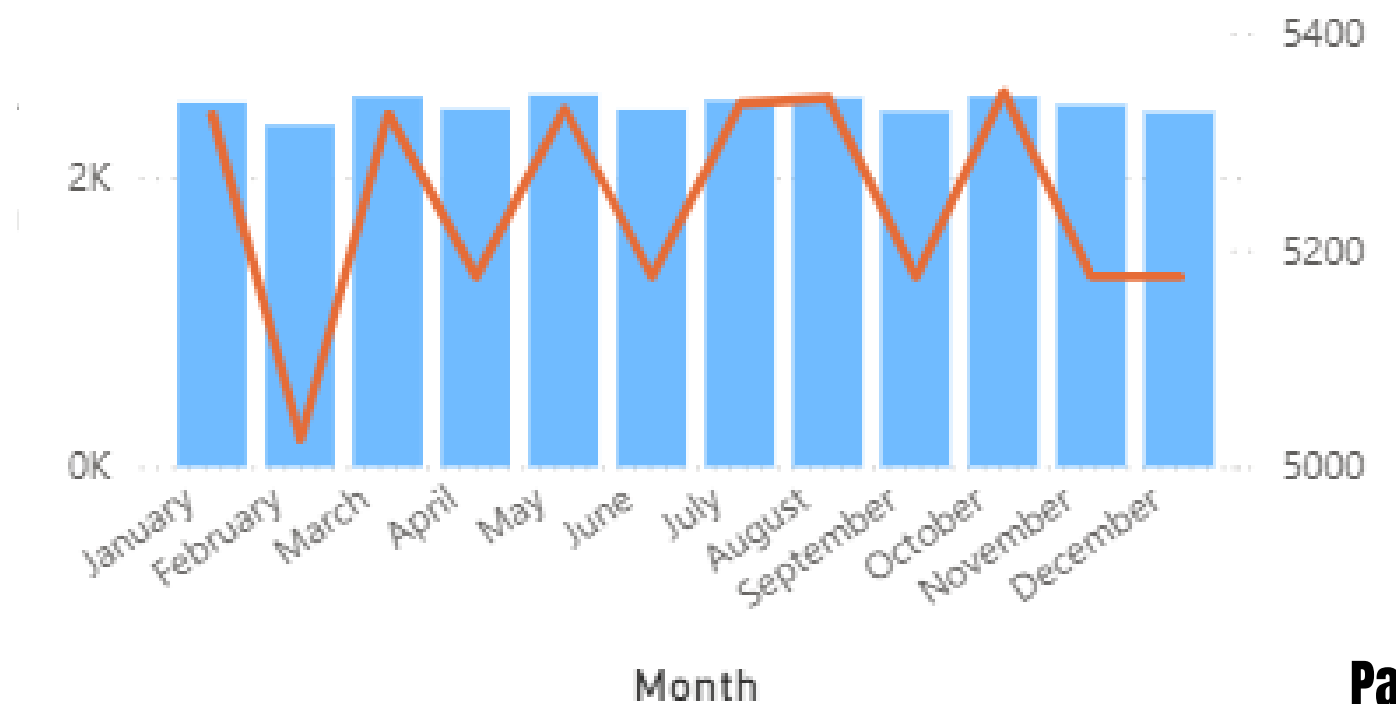
- ▶ IOMT dashboard checks device health
- ▶ Ensures machines work properly
- ▶ Helps management:
 - Digital gaps
 - Management risks
- ▶ Helps in strategic planning
- ▶ There is also Bed occupancy dashboard.

Count of Security IoMT_Enabled



Resource Balance: Bed Occupancy vs. Staffing Trends

Sum of Bed_Occupancy_Rate ● Sum of Staff_On_Duty



SWOT ANALYSIS

This analysis highlights both internal inefficiencies and external pressures.

STRENGTHS

Large patient base provides extensive clinical data, useful for research and improving healthcare services.

WEAKNESSES

Fragmented IT systems and incomplete digital implementation lead to inefficiencies and data gaps.

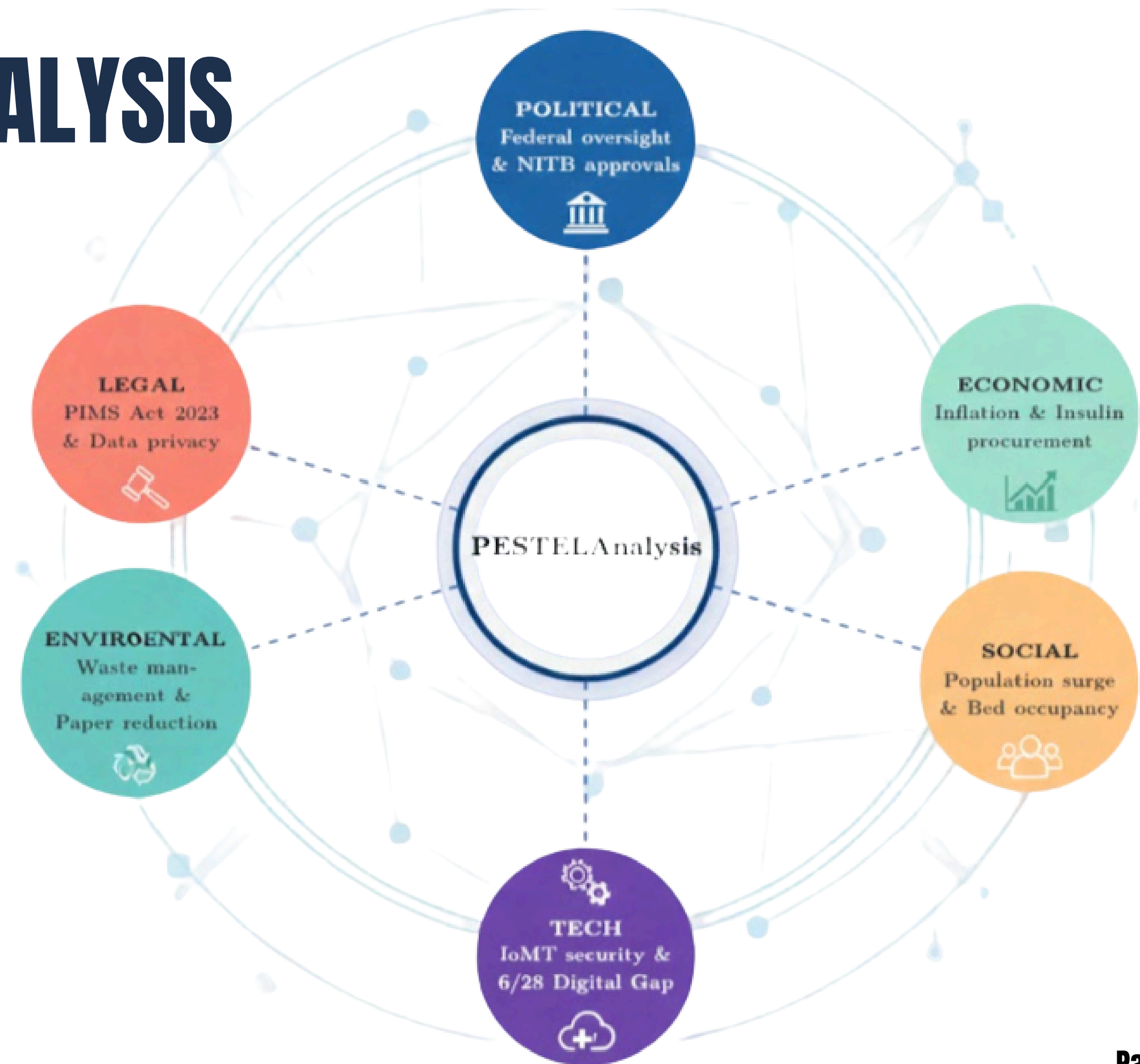
OPPORTUNITIES

Government initiatives for universal health records can modernize operations and improve patient care.

THREATS

Parliamentary scrutiny and risks from medicine stockouts can affect service delivery and reputation.

PESTEL ANALYSIS



POTERS

FIVE FORCES ANALYSIS

Threat of New Entrants

Entering the healthcare sector requires heavy capital investment, strict government approvals, and compliance with medical regulations, making it difficult for new hospitals to enter and compete with PIMS.

Supplier Power

PIMS relies on specialized suppliers such as pharmaceutical companies, and IT vendors. Since there are limited alternatives for these critical supplies, suppliers have greater bargaining power.

Buyer Power

Patients have limited public healthcare options, which reduces their bargaining power. However, expectations for quality care and services still influence PIMS's operations, keeping buyer power at a moderate level.

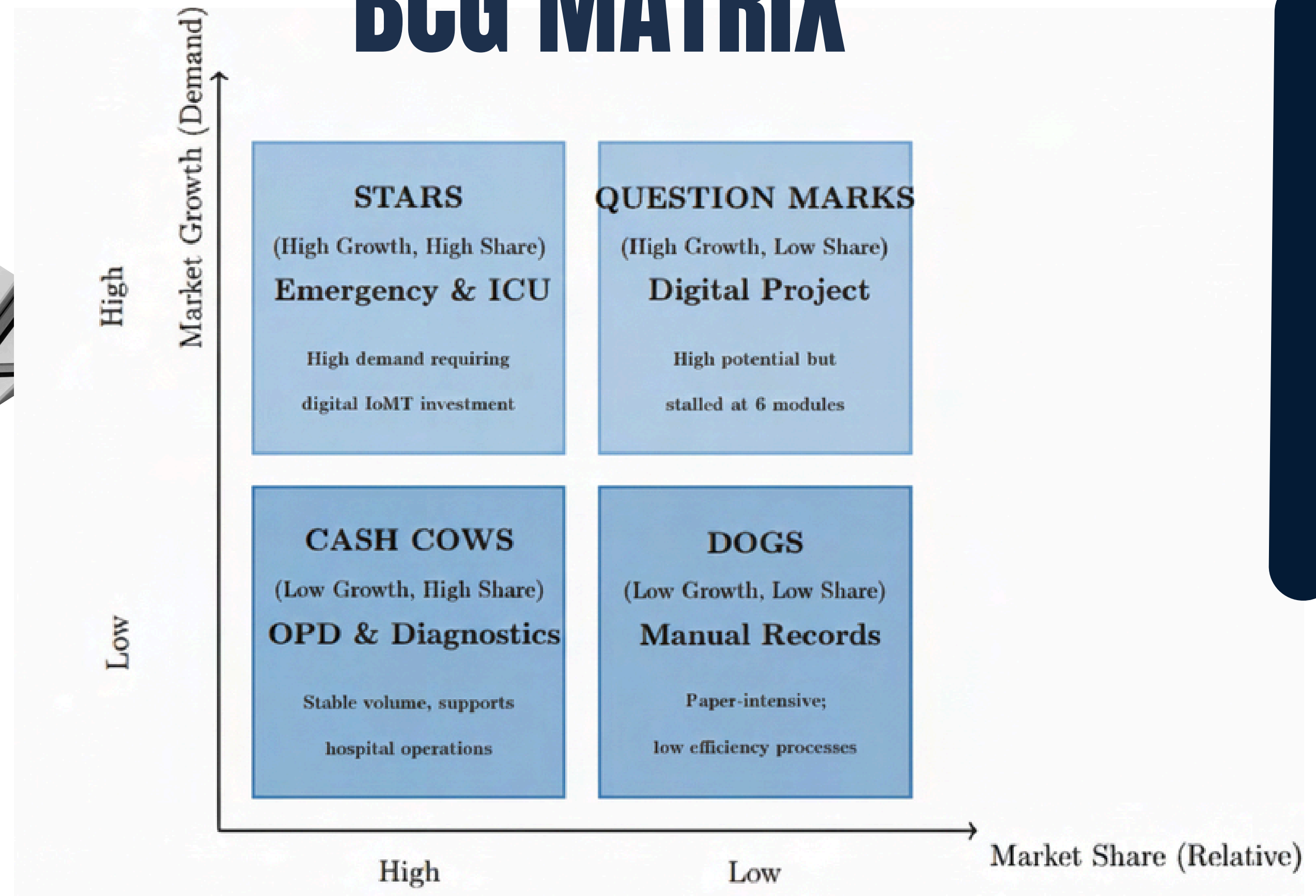
Threat of Substitutes

Private hospitals and clinics act as substitutes by offering faster services and advanced facilities. While they are more expensive, they still attract patients who can afford them.

Competitive Rivalry

PIMS faces competition mainly for government funding, skilled medical staff, and technological resources rather than direct patient rivalry.

BCG MATRIX



LEADERSHIP & OPERATIONS MANAGEMENT

- ▶ Coordination issues reduce staff motivation
- ▶ Democratic leadership is recommended
- ▶ BI supports:
 - Just-in-Time inventory
 - Better resource use
- ▶ Reduces waste and delays

PROJECT MANAGEMENT APPROACH

- ▶ Digital transformation divided into phases:
 - Planning
 - Integration
 - Deployment
- ▶ Gantt chart helps track progress
- ▶ Ensures completion of remaining 22 modules



RISK MANAGEMENT

► Major Risks

- Digital fragmentation
- Medicine shortages
- Internal conflicts
- Data security issues

► Solutions

- Cloud systems
- Automated alerts
- Better coordination
- Secure networks

CONCLUSION

- ▶ PIMS has strong infrastructure.
- ▶ Main issue is partial digital implementation
- ▶ Business Intelligence improves visibility
- ▶ Integration of management & technology is essential
- ▶ Digital transformation improves patient care